Progressive Education Society's
MODERN COLLEGE OF ENGINEERING, Pune -05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
MCA Department

PRACTICAL SUBMISSION RECORD- A.Y. 2024-25

Class: SYMCA Div: B **Semester: III**	**Course Code: MCA01604** **Course Name: Data Science Laboratory**	**Batch: S1**
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CO No: CO605.2	**Assignment No: 08**	

Title: Use the Apriori algorithm on the grocery dataset with minimum support to 0.001 and minimum confidence of 0.8 indicate the top 5 association rules that are generated and highlight the strong ones, sort them by confidence

Code:

```
# --- Step 1: Import necessary libraries ---
import pandas as pd
from mlxtend.frequent_patterns import apriori, association_rules
from mlxtend.preprocessing import TransactionEncoder

# --- Step 2: Data Loading and Preparation ---
filename = 'groceries.csv'
dataset = []
try:
    df = pd.read_csv(filename)

    df = df.drop(columns=df.columns[0])

    for index, row in df.iterrows():
        transaction = row.dropna().tolist()
        dataset.append(transaction)

except FileNotFoundError:
    print(f"Error: '{filename}' not found. Please make sure the file is in the same directory as the script.")
    exit()
except Exception as e:
    print(f"An error occurred: {e}")
    exit()

te = TransactionEncoder()
te_ary = te.fit(dataset).transform(dataset)
df_encoded = pd.DataFrame(te_ary, columns=te.columns_)

# --- Step 3: Apply the Apriori algorithm with your specified support ---
min_support_threshold = 0.001
```

```
frequent_itemsets = apriori(df_encoded, min_support=min_support_threshold, use_colnames=True)
print(f"\nFound {len(frequent_itemsets)} frequent itemsets with support >= {min_support_threshold}.")

# --- Step 4: Generate association rules with your specified confidence ---
min_confidence_threshold = 0.8
rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=min_confidence_threshold)

# --- Step 5: Sort the rules and display the top 5 ---
rules_sorted = rules.sort_values(by='confidence', ascending=False)
print("\n--- Top 5 Association Rules (sorted by confidence) ---")
print(rules_sorted.head(5))
if rules.empty:
    print("No rules found for the given support and confidence thresholds.")
```

Output:

```
Found 13492 frequent itemsets with support >= 0.001.

--- Top 5 Association Rules (sorted by confidence) ---
                                    antecedents          consequents  ...  certainty  kulczynski
376  (oil, root vegetables, other vegetables, yogurt)        (whole milk)  ...        1.0    0.502786
378    (oil, root vegetables, yogurt, tropical fruit)        (whole milk)  ...        1.0    0.502189
360      (whole milk, tropical fruit, ham, pip fruit)  (other vegetables)  ...        1.0    0.502890
371        (whole milk, rolls/buns, soda, newspapers)  (other vegetables)  ...        1.0    0.502627
51       (whipped/sour cream, brown bread, pip fruit)  (other vegetables)  ...        1.0    0.502890

[5 rows x 14 columns]
```